

DISCELL

Author One¹

Author Two²

¹Affiliation One ²Affiliation Two

Abstract

[TODO: abstract]

1 Introduction

Imaging-based spatial transcriptomics measures thousands of genes in every cell of a tissue section, in place (Janesick et al., 2023), so a cell’s expression can be read together with the cells around it. The counts assigned to a cell, however, come from three sources. The first is the cell’s *intrinsic state*: its type, its cell-cycle phase and whatever else it would carry anywhere in the tissue. The second is its *response* to its niche: programmes that change with who its neighbours are and what tissue surrounds it. The third is *spill-over*: transcripts of neighbouring cells assigned to it, because segmentation of densely packed tissue is imperfect. On this platform a typical cell carries about a tenth of its counts from neighbours (Yang et al., 2025).

The response and spill-over are confounded at first order. A cell that receives its neighbours’ transcripts acquires their markers; a cell that responds to its neighbours shifts its programmes in a way that their types predict. Both grow with the share of a neighbouring type (Bilous et al., 2026), and spill-over alone produces apparent cell-cell interaction signals (Mitchel et al., 2026). Where the responding genes are also markers of the neighbours, as for many ligands and receptors, the two leave the same signature. They differ in form (one adds a neighbour’s profile, the other shifts the cell’s own programmes), but from the counts alone the split between them is set by the modelling assumptions, and a flexible decoder can mimic either. Every claim that a spatial effect on expression is real therefore rests, usually silently, on how much of the neighbour-predictable signal was attributed to spill-over.

Related work. Classical models split a gene’s variance into spatial, intrinsic and environmental parts (Svensson et al., 2018; Arnol et al., 2019) or regress expression on niche covariates (Cable et al., 2022; Tanevski et al., 2022), and treat contamination separately, before analysis (Young and Behjati, 2020; Fleming et al., 2023; Petukhov et al., 2022; Mitchel et al., 2026). Among generative models of the cell, each models only one side of the confound (Table 1). On the spill-over side, resolVI (Ergen and Yosef, 2026) writes a cell’s counts as a mixture of its own decoded expression, its neighbours’ expression decoded by the same network, and a background, with the mixing weights estimated per cell. It has no niche term in the cell’s own expression, and it separates the components by expecting true expression to be low-dimensional. A niche response is low-dimensional too, and the part of it that moves a cell toward its neigh-

Table 1: The closest generative models, on the properties this paper turns on. *Spill-over*: misassigned transcripts are part of the generative model. *Invariance*: the regulariser that keeps the intrinsic latent free of the niche. *Intervention*: expression generated under an explicit change of the neighbourhood. ✓ yes, × no. ^aSIMVI instead estimates a spatial treatment effect.

Model	Niche	Spill-over	Invariance	Intervention
resolVI	×	estimated per cell	×	×
SIMVI	✓	×	independence, marginal adversarial, given type	× ^a
MintFlow	✓	×	adversarial, marginal adversarial, given type	delete, replace cells
Cellina	✓	×	adversarial, marginal adversarial, given type	swap neighbours
DISCELL	✓	one fraction, swept	adversarial, given type	relocate a type

bours’ profile acts like spill-over, so that expectation alone does not separate the two. Our spill-over term is resolVI’s neighbour mixture with the weights fixed. On the niche side, NCEM (Fischer et al., 2023) regresses a cell’s expression on its neighbourhood given cell-type labels, including a conditional-VAE variant with a per-cell latent. SIMVI (Dong et al., 2025) splits a cell into an intrinsic latent and a spatially induced one, whose prior is a regression on the neighbours’ intrinsic latents, the closest analogue of our response prior, with an independence penalty applied to the intrinsic latent. MintFlow (Akbarnejad et al., 2025) conditions the intrinsic latent on cell type and keeps neighbour-type composition out of it with type-specific adversarial critics, the type-conditional invariance we also use. Cellina (Moeed et al., 2026) encodes an extrinsic latent from the neighbours’ expression, removes spatial-domain labels from the intrinsic latent adversarially, and predicts expression under a swapped neighbourhood. Celcomen (Megas et al., 2026) and NicheCompass (Birk et al., 2025) model inter-cellular gene programmes without a per-cell split. None of these models spill-over, and MintFlow and Cellina name segmentation errors as a limitation. SIMVI and MintFlow state identifiability results for their latents, but no method states what the counts cannot identify when a response and spill-over compete for them.

Our approach. DISCELL models both sides. A cell’s expected composition is a mixture of its own expression, built from an intrinsic state $\mathbf{z}_i$ and a response $\mathbf{w}_i$ whose prior depends on a descriptor of

the niche, and a fraction κ of its neighbours’ expression (Section 2.1). The counts bound κ from above but never from below (Proposition 1), so we do not estimate it. We refit the model over a grid of κ and report, for each finding, its *breakdown point* κ^* : the smallest spill-over fraction that explains the finding away (Definition 1). This is a sensitivity analysis in the tradition of tipping points and partial identification (Rosenbaum, 2002; Manski, 2003), with κ as the sensitivity parameter. Neither ingredient is new alone: the spill-over term is resolVI’s mixture with fixed weights, and adversarial invariance given type is MintFlow’s. What is new is putting a niche response and spill-over into one model, which creates the confound, and the sweep, which handles it.

Contributions.

- **A generative model in which a niche response and spill-over compete for the same counts** (Section 2.1).
- **Conditional invariance with a held-out probe, as an evaluation protocol.** A type-conditional adversary keeps neighbour composition and an image-derived niche descriptor out of $\mathbf{z}$; a probe trained after the fit measures the residual niche signal against a within-type permutation floor, for every method on the same cells (Section 2.3).
- **A spill-over sweep with breakdown points**, transferable to any model with a spill-over term, and with teeth: **[PENDING RUNS: breakdown-gaps queue: it breaks three findings, while the others hold to $\kappa = 0.4$, four times the typical misassignment]** (Section 2.4).
- **An evaluation on four sections** of ovarian cancer, lung cancer and paediatric lung, plus a serial section held out whole, against resolVI, SIMVI, MintFlow and Cellina, with recovery checks on simulated sections (Section 3). Code: **[URL to be added]**.

2 Model

DISCELL writes a cell’s counts as the sum of its intrinsic state, its response to the niche and spill-over from its neighbours; the first two are latent, the third is a fixed mixture whose size κ is swept (Figure 1).

2.1 Generative model

For each of N segmented cells i on one section we observe a count vector $\mathbf{x}_i$ over G genes with total ℓ_i ,

a lineage label $t_i \in \{1, \dots, K\}$, its position and the morphology image around it. Two cells are neighbours if their Voronoi cells share a face (Delaunay adjacency), with edges longer than $40 \mu\text{m}$ pruned; $\mathcal{N}_i$ is the neighbour set of i . The model is

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (1)$$

$$\mathbf{w}_i \mid \mathbf{c}_i, t_i \sim \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (2)$$

$$\boldsymbol{\rho}_i = \text{softmax} \left(\underbrace{a(\mathbf{z}_i)}_{\text{intrinsic}} + \underbrace{\mathbf{B}\mathbf{w}_i}_{\text{response}} \right), \quad (3)$$

$$\bar{\boldsymbol{\rho}}_i = \sum_{j \in \mathcal{N}_i} \beta_{ij} \boldsymbol{\rho}_j, \quad (4)$$

$$\mathbf{p}_i = (1 - \kappa) \boldsymbol{\rho}_i + \underbrace{\kappa \bar{\boldsymbol{\rho}}_i}_{\text{spill-over}}, \quad (5)$$

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i). \quad (6)$$

Here $\boldsymbol{\rho}_i$ is the cell’s own, *clean* composition over genes. Its intrinsic part decodes $\mathbf{z}_i \in \mathbb{R}^{d_z}$; its response part is linear in $\mathbf{w}_i \in \mathbb{R}^{d_w}$, so that column k of the loadings $\mathbf{B}$ is a gene programme and w_{ik} is how far the cell has moved along it. The decoder has no type input: the only route from identity to expression runs through $\mathbf{z}_i$, which therefore carries the type together with what the label discretises away, such as cell-cycle phase. The prior mean $m_\psi(\mathbf{c}_i, t_i)$ is the response expected of a cell of type t_i in niche $\mathbf{c}_i$; the prior scale is fixed at $\sigma_w = 1$, which pins the scale of $\mathbf{w}$ against that of $\mathbf{B}$.

The niche descriptor. $\mathbf{c}_i$ is a deterministic function of the data, shared by co-located cells and containing no latent:

$$\mathbf{c}_i = \text{GATv2}(\text{embed}(t_i); \{t_j\}_{j \in \mathcal{N}_i}) \oplus \Phi_i \oplus \mathbf{1}[\mathcal{N}_i = \emptyset]. \quad (7)$$

The first part is one graph-attention layer (Brody et al., 2022) over the neighbours’ types, queried with the cell’s own type, which amounts to a learned, type-specific reweighting of the neighbour-type composition $\mathbf{y}_i$. The second, Φ_i , is the embedding by a frozen pretrained image model (Shaban et al., 2025) of a $128 \mu\text{m}$ morphology crop around the cell, in which a disc of radius $25 \mu\text{m}$ around the cell is set to zero, so that it describes the surrounding tissue and not the cell. No distances enter, so that the attention cannot rediscover the spill-over kernel and merge the two mechanisms the model separates (Supplement S2.2).

Spill-over. The influx $\bar{\boldsymbol{\rho}}_i$ is the neighbours’ clean compositions averaged with a fixed kernel, $\beta_{ij} \propto f_{ij} \exp(-d_{ij}/\tau)$, normalised over $\mathcal{N}_i$, where d_{ij} is the distance between centroids, f_{ij} the length of the shared Voronoi face and $\tau = 20 \mu\text{m}$. The kernel

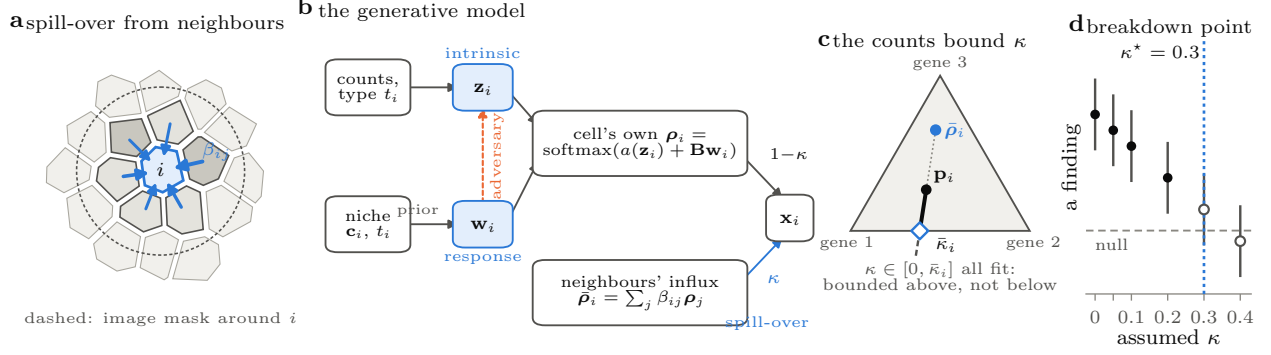


Figure 1: DISCELL in four panels. **a**, Spill-over: a cell (blue) receives transcripts from its neighbours in the pruned Delaunay graph, in proportion to the kernel weights β_{ij} (arrow widths); the niche descriptor $\mathbf{c}_i$ reads the neighbours’ types and the surrounding tissue image, with the cell itself masked out. **b**, The generative model (Equations (1) to (6)): the intrinsic state $\mathbf{z}_i$ and the response $\mathbf{w}_i$, whose prior is set by the niche and the type, decode to the cell’s own composition $\boldsymbol{\rho}_i$, which is mixed with the neighbours’ compositions at the fixed fraction κ . An adversary keeps the niche out of $\mathbf{z}_i$ given the type. **c**, Why κ is swept (Proposition 1): on the simplex of three genes, each assumed κ implies a clean composition on a line running from the observed composition $\mathbf{p}_i$ away from the neighbours’ influx $\bar{\boldsymbol{\rho}}_i$; every κ fits until a gene’s share would turn negative at $\bar{\kappa}_i$, so the data bound κ from above only. Schematic. **d**, The breakdown point (Definition 1): a finding is refitted at every κ of the grid, and κ^* is the smallest κ at which its interval reaches its null value. Schematic.

is computed once and not learned, unlike the per-cell neighbour weights of resolVI (Ergen and Yosef, 2025). The spill-over it represents has a specific form: one hop, one fraction for the whole section, and no gene dependence; an isolated cell receives none (Supplement S1.1).

2.2 Inference and objective

Both latents have amortised diagonal-Gaussian posteriors. The encoder of $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ reads the cell’s library-normalised log counts, its log total and its type, and never the niche. The posterior $q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i)$ reads the niche and also the cell, since $\mathbf{z}_i$ and $\mathbf{x}_i$ together carry evidence about $\mathbf{w}_i$ that the niche alone does not. Writing $r_i(\mathbf{w}) = \ell_i^{-1} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \mathbf{w})$ for the reconstruction per count, the training objective, maximised, is

$$J = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left[r_i(\mathbf{w}_i) + \omega r_i(\tilde{\mathbf{w}}_i) - \alpha_a \text{Adv}_i - (1 + \omega) \alpha_z \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i)) - \alpha_w \text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i)) \right], \quad (8)$$

over a batch $\mathcal{S}$ of cells. With $\omega = 0$ and $\alpha_a = 0$ this is an evidence lower bound, scaled by depth so that the weights transfer between sections of different

depth. The second term is the *intrinsic path*: the same decoder evaluated with $\tilde{\mathbf{w}}_i$ drawn from the prior instead of the posterior, a second bound on the same evidence (Slavutsky et al., 2026) that asks $\mathbf{z}_i$ to explain the cell given only the response *expected* in its niche. It keeps the type out of $\mathbf{w}$: refitted with $\omega = 0$, the response’s agreement with type rises on all four sections (Supplement S8.2). The last term is the adversary of Section 2.3. The neighbours’ $\boldsymbol{\rho}_j$ enter $\bar{\boldsymbol{\rho}}_i$ under a stop-gradient, as in resolVI (Ergen and Yosef, 2025), and batches are contiguous spatial tiles with a two-hop halo (Supplement S3.3). We use $d_z = 20$, $d_w = 6$, $\omega = 1$, $\alpha_z = \frac{1}{2}/\bar{\ell}$ with $\bar{\ell}$ the section’s median total, and $\alpha_w = 0.1$; the remaining settings are in Supplement S3.1.

2.3 Conditional invariance

The architecture keeps the niche out of the encoder of $\mathbf{z}$, but not out of the counts it reads. The target is that, within a type, the intrinsic state says nothing about the niche:

$$I(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] | t_i) = 0. \quad (9)$$

It is conditional on the type because cell types cluster in space: “I am a T cell” predicts “my neighbours are T cells”, and an intrinsic state blind to the niche unconditionally would have to discard its type (Zhao and Gordon, 2019). It covers the image too, so that

a niche defined by morphology alone cannot hide in $\mathbf{z}$. Two heads with parameters ξ predict, from the posterior mean $\mu_{z,i}$ and the type, the neighbour-type composition $\mathbf{y}_i$ and a soft membership e_i^Φ of Φ_i in K image niches; the encoder is penalised by their skill beyond knowing the type alone,

$$\begin{aligned} \text{Adv}_i = \lambda_y [& \text{CE}(\mathbf{y}_i, \bar{\mathbf{y}}(t_i)) - \text{CE}(\mathbf{y}_i, \hat{\mathbf{y}}_\xi(\mu_{z,i}, t_i))] \\ & + \text{CE}(e_i^\Phi, \bar{\Phi}(t_i)) - \text{CE}(e_i^\Phi, \hat{e}_\xi^\Phi(\mu_{z,i}, t_i)), \end{aligned} \quad (10)$$

where $\bar{\mathbf{y}}(t)$ and $\bar{\Phi}(t)$ are the mean targets of type t and $\lambda_y = 3$ weights the composition, the channel of the spill-over confound. The heads maximise Adv_i with their own optimiser, the encoder minimises it, and it is zero at the optimum. Type-conditional adversarial invariance of an intrinsic latent against neighbour-type composition was introduced by MintFlow’s per-type critics (Akbarnejad et al., 2025); what we add is the image target and an independent test of the result.

The held-out probe. A training adversary that is defeated proves nothing (Elazar and Goldberg, 2018), so invariance is graded by a *fresh* predictor of each target block b (the composition, and the leading principal components of Φ) from $(\mu_{z,i}, t_i)$, trained on spatially separate training tiles and scored on held-out tiles. Its held-out gain over the type mean $\bar{v}_k(t)$, per component k of the block,

$$G_b = \frac{1}{|b|} \sum_{k \in b} \frac{1}{2} \log \frac{\mathbb{E} (v_k - \bar{v}_k(t))^2}{\mathbb{E} (v_k - \hat{v}_k(\mu_z, t))^2}, \quad (11)$$

estimates the predictive information from $\mathbf{z}$ to the block given the type (Xu et al., 2020). Its value with μ_z permuted among cells of the same type is the chance floor. The **residual niche signal** is the excess of G_b over that floor, reported as $1 - e^{-2 \text{ excess}}$: the share of the block’s within-type variance that the probe explains beyond chance. We fit a linear (ridge) and a nonlinear (small perceptron) probe, each against its own floor, and compare with a fit without the adversary ($\alpha_a = 0$). A probe at its floor bounds the dependence it can detect; it does not certify independence. Checkpoints are selected on held-out reconstruction while the agreement of $\mathbf{z}$ with type (normalised mutual information of a k -means clustering of μ_z with t) stays within 90% of its running maximum (Supplement S3.1).

2.4 What is identified

Spill-over and response both make a cell resemble its neighbours, and a flexible model can fit either.

The counts constrain the spill-over fraction in one direction only. Removing more and more of the neighbours’ profile from a cell eventually drives some gene’s share below zero, which no clean profile can have: this bounds κ from above. Every smaller κ yields a valid clean profile that reproduces the data exactly, so there is no lower bound (Figure 1c).

Proposition 1. *Hold the neighbours’ clean compositions, and hence $\bar{\rho}_i$, fixed, and let $\mathbf{p}_i = (1 - \kappa)\rho_i + \kappa\bar{\rho}_i$ with ρ_i in the simplex. A fraction $\kappa' \in [0, 1]$ reproduces $\mathbf{p}_i$ with a clean composition in the simplex if and only if $\kappa' \leq \bar{\kappa}_i = \min_{g: \bar{\rho}_{ig} > 0} p_{ig}/\bar{\rho}_{ig}$. The true κ satisfies $\kappa \leq \bar{\kappa}_i$, with equality if and only if $\rho_{ig} = 0$ for some gene g with $\bar{\rho}_{ig} > 0$.*

The proof is in Supplement S2.6. Even with the neighbours’ clean profiles known, then, the counts bound κ only from above, and pin it only through a gene that the cell cannot express itself. With finite counts the bound is soft; we do not compute it on our sections. In simulation, recovery of planted states and programmes barely changes with the assumed κ (Section 3.2), which is the same non-identifiability seen in practice. Given κ , the response is identified at most up to a rotation of its programmes and a per-type offset, so we report only rotation-invariant summaries and within-type contrasts (Supplement S1.5).

The sweep. We therefore do not estimate κ . We fit the model independently at each $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$, with three seeds per value and every other weight held fixed, and check the held-out probe at each value, so that a change in a readout is not a change in the invariance. This is a sensitivity analysis with κ as the sensitivity parameter, in the tradition of tipping-point and E-value analyses of unmeasured confounding (Rosenbaum, 2002; VanderWeele and Ding, 2017): the fractions compatible with the data form an interval from zero (Manski, 2003), and the question asked of each finding is how far into it the finding survives.

Definition 1 (Breakdown point). For a signed contrast r , let $I_\kappa(r)$ be the interval of its mean over seeds at spill-over fraction κ . The contrast holds at κ if $I_\kappa(r)$ excludes zero and every seed keeps the sign the contrast has at $\kappa = 0$. Its *breakdown point* $\kappa^*(r)$ is the smallest grid value at which it does not hold. If it does not hold at $\kappa = 0$, r is no finding; if it holds at every grid value, $\kappa^*(r)$ is reported as exceeding the top of the grid.

The interval comes from a spatial block bootstrap over tiles, Bonferroni-adjusted over the contrasts claimed on a section. A finding with $\kappa^* > 0.4$ cannot

be explained by spill-over of the modelled form up to four times the typical misassignment of about a tenth of a cell’s transcripts (Yang et al., 2025); spill-over of another form (longer-ranged, gene-specific, depth-dependent) is not ruled out. Readouts shrink with κ by construction, so only sign and interval are read. A contrast without a sign at $\kappa = 0$, because it is zero there by construction or measures decoding rather than spill-over removal, is shown as a trajectory instead (Supplement S8.1). Nothing in the protocol is specific to DISCELL: any model with a spill-over term at a fixed fraction can attach a breakdown point to any spatial readout.

3 Experiments

3.1 Setup

Sections and labels. We fit DISCELL to four sections from one imaging-based assay with a panel of about five thousand genes (Janesick et al., 2023): ovarian cancer fixed in formalin and embedded in paraffin (FFPE), our primary section (10x Genomics, 2024a); lung cancer, FFPE (10x Genomics, 2024c); fresh-frozen (FF) ovarian cancer, with about eight times the counts per cell (10x Genomics, 2024b); and one core of a paediatric lung tissue microarray (TMA) (Williams-Katek et al., 2026). They hold 69,000 to 1.16 million cells (Supplement S4.1). A serial section of the TMA core, on a second slide, is never fitted; it is read only through the models fitted to the core. Cells carry lineage labels (curated on the ovarian FFPE section and the TMA, marker-annotated clusters elsewhere), in which finer states and locations are merged into their lineage; unlabelled cells train as their own class and are never scored.

Splits, seeds and the operating point. 15% of each section’s contiguous spatial tiles are held out, and every readout is taken on held-out cells. Every configuration is fitted with three seeds, each redrawing the split, and reported as the mean with the range over seeds. One configuration, with α_z scaled to depth, serves every section. Single fits are read at $\kappa = 0.1$, the typical misassignment of about a tenth of a cell’s transcripts (Yang et al., 2025). This is an operating point, not an estimate: every claimed contrast below is re-examined across the sweep in Section 3.7.

Comparison methods and fairness. We compare with resolVI (Ergen and Yosef, 2025), a single-latent model of spill-over and our reference for the

intrinsic side, and with SIMVI (Dong et al., 2025), MintFlow (Akbarnejad et al., 2025) and Cellina (Moeed et al., 2026), which split each cell into an intrinsic and a context latent. All are scored by the same readouts of their intrinsic latent on our held-out cells. The comparison is asymmetric in two opposite directions: DISCELL’s weights were calibrated on these sections (Supplement S8.2), while the comparison methods ran at their published defaults and were trained, as their software is designed, on all cells, our held-out cells included. Cellina removes a spatial-domain label from its intrinsic latent; our sections have none, so it is also shown with our niche label as its domain, its best case on our probe.

3.2 Simulation

We simulate sections from the generative model of Section 2.1: 6,000 cells of eight spatially clustered types, a response linear in the neighbour composition, and spill-over planted at $\kappa = 0$ or 0.2, three worlds each, each fitted with three seeds (Supplement S7.1, Table S20).

Recovery. The final configuration recovers what was planted. The intrinsic state agrees with the planted type (NMI 0.80 to 0.96), as well as clusters of the counts or better in 17 of 18 fits; the response correlates with the planted one (first canonical correlation 0.83 to 0.95); the loadings span the planted programmes better than random subspaces in every fit; and the intrinsic state carries almost none of the planted niche (within-type R^2 of composition at most 0.04).

The adversary makes the response recoverable. Without it, the response correlates with the planted one at only 0.29 to 0.91, and the niche moves into the intrinsic state (composition R^2 0.11 to 0.19): a cell’s resemblance to its neighbours can then be explained by $\mathbf{z}$ as well as by $\mathbf{w}$.

A wrong κ costs little, so the fit cannot find the right one. On the worlds planted at $\kappa = 0.2$, fits at any assumed κ from 0 to 0.4 recover the planted quantities about equally well (Figure S15). Recovery thus does not hinge on guessing κ , and it offers no handle on κ either: the non-identifiability of Proposition 1, seen in practice, and the reason we sweep κ . The encoder’s posterior mean is measurably off the exact one, more so at the final configuration than in development (Supplement S7.3).

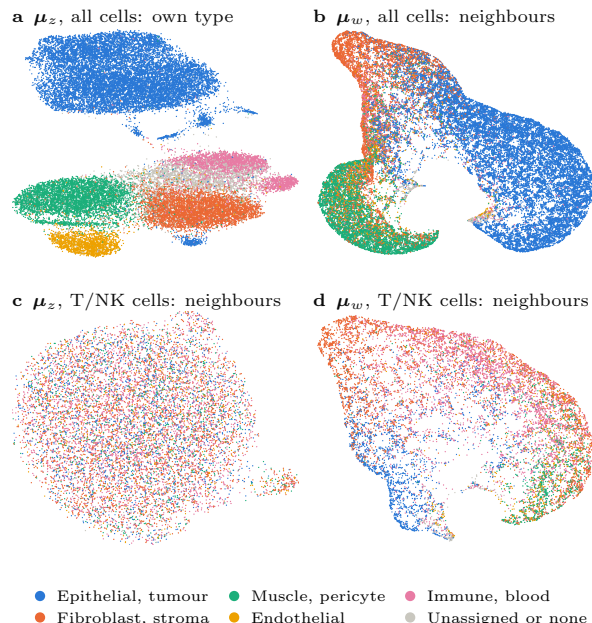


Figure 2: The two latents on the ovarian FFPE section (one seed; UMAP of the posterior means). **a**, μ_z of 30,000 random cells, coloured by the cell’s own lineage. **b**, μ_w of the same cells, coloured by the lineage that dominates the cell’s neighbours. **c**, μ_z of T/NK cells, the large type with the most mixed neighbourhoods (chosen by that rule), coloured by their neighbours. **d**, μ_w of the same T/NK cells. UMAP can exaggerate separation; the held-out probe measures what c and d show.

A positive control for the breakdown point.

A breakdown point is useful only if spill-over can break a finding. We therefore plant a spatial contrast created by spill-over alone, at a known κ_{true} , beside a genuine response contrast, and sweep both. The prediction, fixed in advance, is that the spill-over contrast breaks down at the first grid value at or above κ_{true} , while the response contrast holds across the grid. **[PENDING RUNS: planted spill-over control: the spill-over-only contrast breaks down at $\kappa^* = \dots$ for $\kappa_{\text{true}} = \dots$, and the response contrast holds to $\kappa = \dots$. If κ^* does not track κ_{true} , say so and present the breakdown point as a lower-power check.]**

3.3 What the Intrinsic State and the Response Carry

Figure 2 shows the split at a glance. The intrinsic state groups cells by their own lineage (a), and the response by the lineage of their neighbours (b). Within

a single type, the T/NK cells, **z** shows no grouping by the neighbours (c), while **w** is ordered by them (d). This is what conditional invariance asks for. Table 2 measures it on every section.

The intrinsic state keeps the type and the cell cycle. On every section **z** agrees with the lineage labels (NMI 0.62 to 0.74) and carries the cell cycle, a within-type state the labels do not encode: on the most strongly cycling tenth of held-out cells, **z** explains 19 to 76% of the variance of the cycle scores and **w** at most 1% (Table 2). The cycle score is a proxy computed from the same segmented counts, which spill-over from cycling neighbours also raises; the asymmetry shows where the model puts this axis, not that the score is free of spill-over. Read without refitting, the serial section gives what the core gives (NMI 0.61 against 0.62; cycle R^2 of **z** 0.50 on both): an out-of-sample replication on a second slide.

The adversary removes most of the niche from **z**.

On the four fitted sections it cuts the residual niche signal (Section 2.3) that a held-out linear probe finds in **z** by 80 to 93%. A nonlinear probe still finds 3.5 to 5.5% of the within-type variance of neighbour composition on every section (Table S6); on the TMA sections, which carry little niche signal even without the adversary, that is more than half of the original amount. A probe-free read agrees: the within-type spatial autocorrelation (Moran’s I) of **z** falls by 25 to 48% (Figure 3d).

The response carries the niche. **w** holds 0.47 to 0.60 nats of information about the niche beyond a within-type permutation floor (Table 2), 85 to 90% of what the two latents carry together (Supplement S6.1). Without the adversary the response nearly vanishes within types and **z** takes up the niche instead (Supplement S6.2): on tissue, as in simulation, the adversary is what makes the split exist.

Held-out sublabels split as designed. The lineage labels merge finer curated classes that the model never saw, each with a known allegiance: a tumour *state* belongs to **z**, a *location* to **w**. On the primary section, probes trained on training tiles and scored on held-out tiles show a double dissociation (Supplement S5.2). Tumour- against stroma-associated fibroblasts are read from **w** with balanced accuracy 0.95 (from **z**: 0.66), and the same split of endothelial cells with 0.90 (0.67). The four tumour states are read from **z** with 0.79 (from **w**: 0.41; chance 0.25). By the call fixed in advance, the proliferative and

Table 2: Model quality on the four sections and the held-out serial section of the TMA core: the mean over three seeds, with the range over seeds in brackets; the 95% intervals are in Table S14. Reconstruction is the held-out log-likelihood in nats per count. Transport is the fraction of the noise ceiling (cell-split reliability) recovered by the mean transport read, over the panels in the trusted tier ([†]seeds with a trusted panel: TMA core, 2 of 3). The same read scored on held-out tiles only is a sensitivity row (Table S15). Cycle R^2 is read on the top decile of the S and G2M score among held-out cells. I(niche; $\mathbf{w}$) is the excess over a within-type permutation floor, in nats. The atlas cosine compares the first programme axis between seeds. The serial section is read through the models fitted on the core (–: not read there). Arrows give the preferred direction; ≈ 0 : expected to be near zero. Unassigned cells are not targets of any read.

Read	Ovarian FFPE	Lung FFPE	Ovarian FF	TMA core	TMA serial
Reconstruction $\uparrow$	–7.193 (–7.232 to –7.148)	–7.254 (–7.267 to –7.243)	–7.320 (–7.329 to –7.312)	–7.219 (–7.223 to –7.213)	–7.237 (–7.244 to –7.232)
NMI of $\mathbf{z}$ with the type $\uparrow$	0.713 (0.701–0.723)	0.736 (0.733–0.739)	0.697 (0.673–0.719)	0.621 (0.617–0.627)	0.611 (0.606–0.614)
Mirror R^2 $\downarrow$	0.048 (0.044–0.051)	0.026 (0.025–0.027)	0.091 (0.084–0.094)	0.028 (0.028–0.028)	0.039 (0.039–0.039)
Cycle R^2 of $\mathbf{z}$ $\uparrow$	0.519 (0.504–0.548)	0.186 (0.177–0.195)	0.757 (0.744–0.768)	0.497 (0.476–0.509)	0.495 (0.494–0.496)
Cycle R^2 of $\mathbf{w}$ (≈ 0)	0.006 (0.003–0.009)	0.003 (0.001–0.006)	0.001 (0.000–0.002)	0.000 (–0.000 to 0.002)	–0.001 (–0.004 to 0.002)
I(niche; $\mathbf{w}$) excess $\uparrow$	0.47 (0.45–0.50)	0.59 (0.58–0.62)	0.60 (0.58–0.62)	0.48 (0.42–0.52)	–
Transport $\uparrow$	0.65 (0.63–0.67)	0.72 (0.68–0.75)	0.67 (0.65–0.68)	0.73 [†] (0.71–0.75)	–
Atlas cross-seed cosine $\uparrow$	0.97 (0.97–0.98)	0.84 (0.80–0.90)	0.82 (0.76–0.90)	0.91 (0.89–0.92)	–

inflammatory states go to $\mathbf{z}$ on all three seeds and the locations to $\mathbf{w}$ on two of three. The exceptions are states that are also places: hypoxic VEGFA⁺ tumour cells are read from both latents, and myofibroblasts on the TMA core better from $\mathbf{w}$. Merging such a state into its lineage moves part of it into the response.

At the operating point the response is type-level. The response and its prior mean $m_\psi(\mathbf{c}_i, t_i)$ read the sublabels alike (balanced accuracy 0.95 and 0.94 for the fibroblasts), and the response’s own posterior adds one part in a thousand or less to held-out reconstruction (Supplement S8.3). A per-cell response planted in simulation, unpredictable from the niche and twice the niche effect, is not detected (Supplement S7.2). At $\alpha_w = 0.1$ the per-cell channel is closed by design: the response describes how a cell *type* shifts with its niche, not how one cell deviates from that shift; the lung section, where it separates some clusters better than its prior mean, is a partial exception.

3.4 What the Response Encodes

Within types, the response’s variation lies in two or three programmes (four or five on the FF section, depending on the seed). Programmes, directions of $\mathbf{w}$ read in gene space through $\mathbf{B}$, are identified only up to rotation (Section 2.4), so we describe each section’s leading programme within one fit (Figure 3) and compare seeds only on its leading genes and hallmark labels.

The response maps territories. Within types, Moran’s I of $\mathbf{w}$ is 0.48 to 0.73, against 0.05 to 0.11 for $\mathbf{z}$ on three sections and 0.35 on the FF section (Figure 3d).

Each leading programme has a recognisable signature. On the ovarian FFPE section the leading programme (70% of the response’s within-type variance) sets *COL11A1*, *MMP11*, *COL10A1* and *INHBA*, genes of tumour-associated, matrix-remodelling stroma [NEEDS A SOURCE], against quiescent-fibroblast (*C7*, *DPT*, *TNXB*) [NEEDS A SOURCE] and inflammatory genes (*IL6*, *PTGS2*). Its territory lies around the tumour: in fibroblasts, macrophages, T/NK and endothelial cells the score rises with the share of tumour cells among the neighbours. The same genes lead it in all three seeds, as do its hallmark labels, epithelial–mesenchymal transition (EMT) and TNF α /NF- κ B signalling. On the lung FFPE section the leading programme sets *CCL19*, *PLVAP*, *ADAMDEC1* and *MMP9*, genes of inflamed stroma [NEEDS A SOURCE], against the alveolar *AGER* and *MME*, with the same two labels in all seeds; on the TMA core, adventitial-fibroblast genes (*MFAP5*, *PI16*, *PDGFD*) against alveolar-fibroblast genes (*CCN5*, *NPNT*, *FAT3*) [NEEDS A SOURCE]. The labels are coarse: EMT, dominated by extracellular-matrix genes, is significant on most programmes, so the genes are the more specific read.

What drives each programme. As the response is close to its prior mean, composition and image

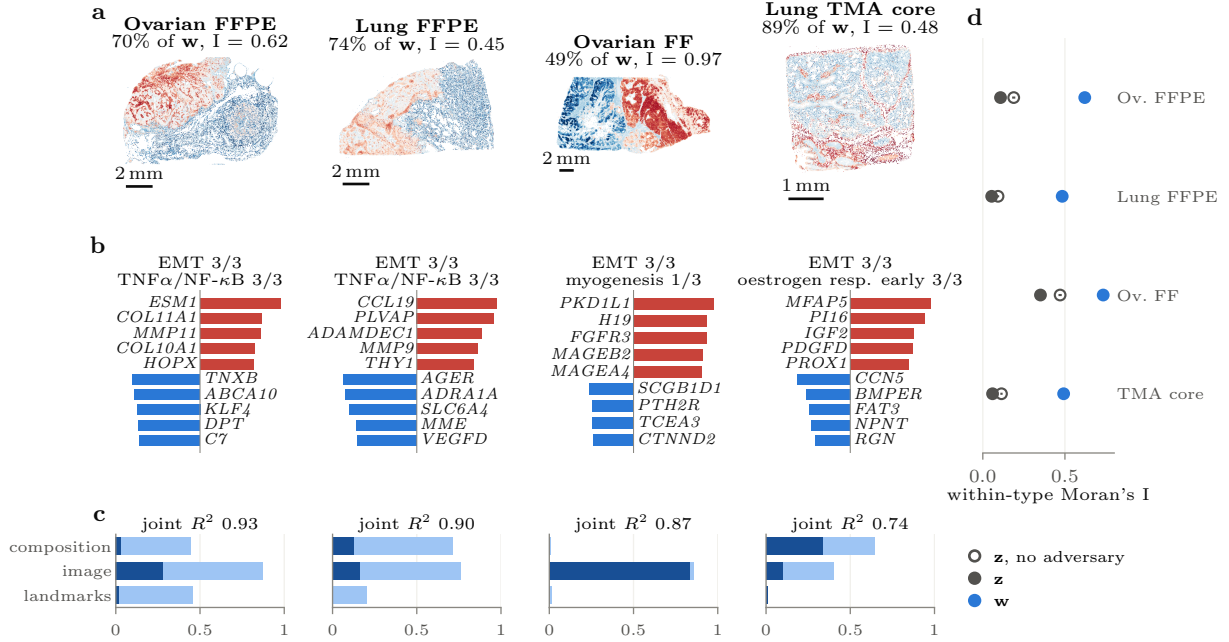


Figure 3: The leading response programme of each section, in one fit. **a**, Territory: each cell’s score, within its type (red high; the FF section is rotated), with the programme’s share of w ’s within-type variance and its within-type Moran’s I . **b**, Signature: the five largest positive and negative gene loadings, scaled, and its two most significant hallmark labels (Liberzon et al., 2015), with the number of seeds in which the matching programme carries each among its top three at $q \leq 0.05$. **c**, Drivers: cross-validated R^2 of the score from neighbour composition, image context and distances to landmarks (vessels, smooth muscle, the tumour–stroma interface, where annotated), alone (light) and unique given the others (dark). **d**, Within-type Moran’s I of z without the adversary (two seeds), and of z and w in the final fits (three seeds): mean and range.

together explain most of each programme (joint R^2 0.74 to 0.93, Figure 3c); what matters is which. The image carries the unique share on the ovarian FFPE section, composition on the TMA core, both on the lung section. Distances to landmarks, not a model input, explain up to 0.46 of the ovarian programme alone but at most 0.02 beyond composition and image, which already reflect them.

The FF section is the least stable. Its leading programme follows the image almost alone (unique R^2 0.83), splits the section into two large regions, and is led by tumour-cell genes such as the cancer-testis antigens *MAGEA4* and *MAGEB2*. A regional difference between tumour areas, seen through the image, explains it as well as a response does, and the model cannot tell the two apart. Across seeds its leading direction matches least well (cosine as low as 0.67, against 0.72 to 0.98 elsewhere), the number of programmes changes, and only its EMT label recurs. We do not interpret it as a niche response.

3.5 Comparison with Other Methods

Every method is scored by the same read-outs on its intrinsic latent, on our held-out cells (Section 3.1). Figure 4 sets two of them against each other: the residual niche signal (Section 2.3) and the within-type cell-cycle state the latent keeps. We use cycle state as the second axis because it is the one read of retained state that needs no label and varies within a cell type; it was fixed for all methods at once, after earlier baseline reads but before the final scoring. Agreement with the type label (NMI) is not a substitute: it rewards label fidelity, which MintFlow and Cellina are trained towards, and on it both score above DISCELL (Table S11).

No method both carries less residual niche signal than DISCELL and keeps more cycle state, on any section. Under a linear probe, DISCELL’s intrinsic state carries the least neighbour composition and the least image information of all methods on every section (on ovarian FFPE, 1.0–1.1% of the within-type

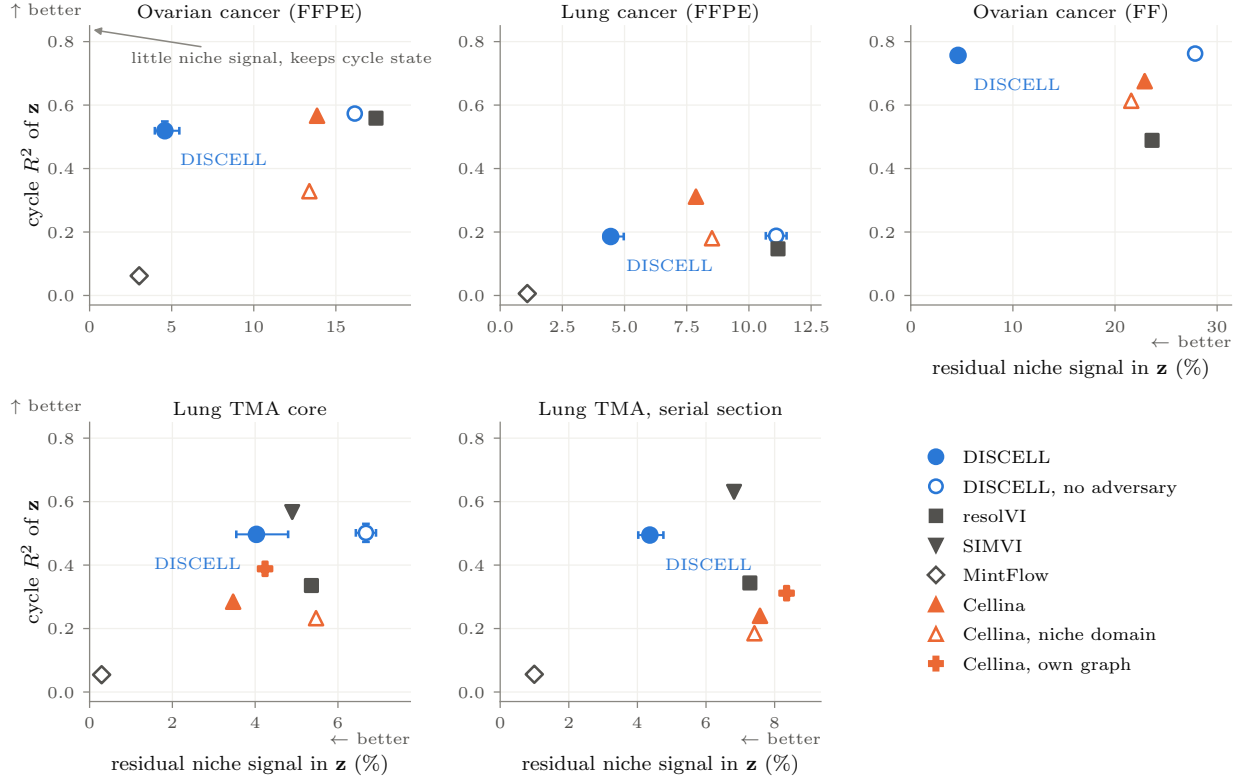


Figure 4: Residual niche signal in the intrinsic latent against the cell-cycle state it keeps, per section. *Horizontal*: share of the within-type variance of neighbour composition that the held-out nonlinear probe reads from each method’s intrinsic latent beyond the permutation floor, in %. *Vertical*: R^2 of the cell-cycle scores from the intrinsic latent on the most strongly cycling tenth of held-out cells. The desirable corner is the top left. DISCELL: mean over three seeds, bars the range; open circle: DISCELL without the adversary. resolVI keeps a single latent and is shown as a reference. Cellina is also shown with our niche label as the domain its adversary removes (its best case on this probe), and on the TMA sections on its own neighbour graph. SIMVI ran out of memory on a 24 GB GPU on the three larger sections, and MintFlow could not be fitted on the fresh-frozen section; both are shown where their whole-section fits exist. The values are in Tables S6 and S11. [PENDING RUNS: MintFlow refits: MintFlow’s points are redrawn from the new fits]

variance of composition, against 4.0% for MintFlow, the next lowest). Under the nonlinear probe of Figure 4, only MintFlow carries less wherever it ran, and its intrinsic latent keeps almost no within-type state (cycle $R^2 \leq 0.06$); on the TMA core, Cellina’s residual is at DISCELL’s lowest seed, with less cycle state. The methods that keep more cycle state carry more residual signal (Cellina on the two FFPE sections, SIMVI on the TMA sections), and on the fresh-frozen section DISCELL is best on both axes. DISCELL’s latent is also the one least predicted by the niche: the within-type R^2 of a linear map from DISCELL’s niche descriptor $\mathbf{c}_i$ to each method’s intrinsic latent (mirror R^2) is lowest for DISCELL on four of the five sections, MintFlow being lower on the

serial one. [PENDING RUNS: MintFlow refits: the MintFlow comparisons in this paragraph are re-scored]

resolVI keeps a single latent for the cell, so it is a reference for a model that does not separate the niche, not a competitor on this axis; its residual is two to five times DISCELL’s on the three whole-slide sections. The result is not reproduced by adding an adversary to another model. Given our niche label as the domain its adversary removes, Cellina’s best case on this probe, its residual does not consistently fall, and it loses agreement with type and cycle state on every section; on its own neighbour graph it keeps less cycle state than DISCELL at a similar or higher residual (Supplement S4.3). The separation comes

Table 3: Relocation, DISCELL against Cellina’s neighbour-rewiring counterfactual, on the same panels, niches and ceilings (DISCELL’s first seed, which supplies the panels; Cellina refitted on the training tiles). *Ceiling*: share of the reproducible between-niche difference recovered, over the panels whose ceiling can be read. *Single cell*: share of the distribution gap to the target niche’s own cells that the moved cells close. *Twin*: how much closer, in expression, each target cell lies to the moved cell nearest it in intrinsic state than to a random moved cell. D: DISCELL; C: Cellina. Higher is better; bold marks the better of the two. ^aA single readable panel. Intervals: Table S17.

Section	Ceiling		Single cell		Twin	
	D	C	D	C	D	C
Ovarian FFPE	0.650	0.731	0.789	0.681	0.398	0.361
Lung FFPE	0.743	0.703	0.826	0.757	0.432	0.435
Ovarian FF	0.677	0.566	0.823	0.602	0.437	0.330
TMA core	0.711	0.899^a	0.814	0.801	0.464	0.284

from the architecture, not from the adversary alone.

Held-out reconstruction is within 0.015 nats per count of the best method on every section and best on the two whole-slide FFPE sections, although the comparison methods were trained on these held-out cells (Table S11). One DISCELL fit, its evaluations included, is up to 2.6 times faster than the fastest comparison method, Cellina, on three of the four sections and slower than it on the fresh-frozen section; it is over a hundred times faster than MintFlow wherever MintFlow could be run (Table S19). Because the result is a sweep of 18 fits (six spill-over fractions, three seeds; Section 2.4), the honest unit is the sweep: at the operating point’s run time it takes at most 2.8 hours on each section where MintFlow ran, less than one MintFlow fit there (9 to 16 hours), and on the TMA core less than one SIMVI fit; on the fresh-frozen section it takes longer than one resolVI fit. **[PENDING RUNS: MintFlow refits: MintFlow’s reconstruction and run times]** The full battery, probe and context-side tables are in the supplement (Tables S6, S11 and S12).

3.6 Relocation

Relocation asks how a cell type’s expression would change if its cells sat in another niche. DISCELL answers it by keeping each cell’s intrinsic state, moving the prior of its response and its spill-over influx to the target niche, and decoding. For each type and pair of niches (a panel), the predicted change in the

type’s mean expression is scored against the observed difference between the two niches, as a share of the part of that difference that is reproducible between two random halves of the cells (the noise ceiling); only panels whose ceiling is high enough to be read count (Supplement S5.1).

Moved to another niche, a cell type’s predicted shift recovers 0.65 to 0.73 of the reproducible between-niche difference (Table 2). The response programmes of Section 3.4 carry most of it: predicted from the programmes alone, the shift recovers 0.52 to 0.64 of the ceiling, and from the spill-over alone 0.21 to 0.39 (the two parts are not additive; Table S17).

Cellina (Moeed et al., 2026) also predicts expression under a rewired neighbourhood, so we score its counterfactual by the same reads on the same panels (Table 3). The comparison is mixed. DISCELL’s moved cells are closer to the target cells’ own distribution on every section, with disjoint intervals on three, and DISCELL leads on all three reads on the fresh-frozen section. Cellina recovers more of the ceiling on the ovarian FFPE section, and on the TMA core on its single readable panel, and is narrowly ahead on the twin read on lung. Most scored cells lie in the model’s training tiles; scored on held-out tiles only, DISCELL’s share of the ceiling moves in both directions across fits, so it shows no systematic in-sample advantage, while Cellina’s falls on both FFPE sections, on lung from 0.70 to 0.49, below DISCELL’s range (Table S15).

Would a plain regression do as well? Regressing each type’s expression shift directly on neighbour composition, from the counts as observed, recovers **[PENDING RUNS: plain-regression reference: share of the ceiling and single-cell read on the same panels]** of the ceiling on the same panels. **[PENDING RUNS: keep one of the two planned sentences once the reference lands]** *If below DISCELL*: modelling spill-over and keeping the intrinsic state apart also makes the relocation more accurate than the niche composition alone predicts. *If level with DISCELL*: the relocation’s accuracy then comes from the niche composition itself, and DISCELL’s advantage lies in the separation, an intrinsic state largely free of the niche and spill-over modelled rather than absorbed, not in relocation accuracy.

3.7 Robustness to Spill-over

Every section was refitted along the sweep of Section 2.4, and Table 4 gives each finding of Sections 3.3 to 3.6 its breakdown point: how much spill-over, assumed in the model, it takes to explain the finding away.

Table 4: Breakdown points κ^* : the smallest assumed spill-over fraction at which a finding stops holding, per section, on the grid $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ with three seeds each. >0.4 : the finding holds at every grid point. Bold: a break. LR: ligand–receptor. ^aThe interval of the seed mean contains 0; ^ba seed has the opposite sign. n/a: no tumour cells on the TMA sections. The serial section is read through the core’s fits. **?: [PENDING RUNS: breakdown gaps, landing tonight; findings per section become 7/7/7/6/6]**. Trajectories and diagnostics: Supplement S8.1.

Finding	Ov. FFPE	Lung FFPE	Ov. FF	TMA core	TMA serial
<i>Biological claims</i>					
Relocation > spill-over part	0.4^b	0.3^b	>0.4	>0.4	?
Relocation > type mean	>0.4	>0.4	>0.4	>0.4	?
Nearest-twin advantage	>0.4	>0.4	>0.4	>0.4	?
LR-gene lean of w	>0.4	>0.4	0.2^a	>0.4	?
Tumour axis in w	>0.4	?	?	n/a	n/a
<i>Allocation checks</i>					
Cycle in z , not in w	>0.4	>0.4	>0.4	>0.4	>0.4
Niche information in w	>0.4	>0.4	>0.4	>0.4	?

The test has teeth. It breaks three findings: the response’s lean toward ligand–receptor genes on the fresh-frozen section ($\kappa^* = 0.2$), and the relocation’s margin over its spill-over part on the ovarian and lung FFPE sections ($\kappa^* = 0.4$ and 0.3 , where a seed changes sign). These are the findings that spill-over itself could most plausibly produce: a gene-specific spill-over would create the lean, and the margin measures what the prediction adds to spill-over. Every other finding holds across the whole grid on every section, up to four times the typical misassignment. **[PENDING RUNS: breakdown gaps: the tumour axis on lung FFPE and ovarian FF and five findings on the serial section; recount the breaks]** The table lists the allocation checks, that **z** keeps the cycle and **w** the niche, apart from the biological claims, because the split favours them by design; the biological claims are the stronger test. Two contrasts first in the graded set were moved to trajectories after reading, for the reasons of Section 2.4: one, between marker pairs, measures decoding, and the other, the relocation’s margin over its programme part, is zero by construction at $\kappa = 0$.

A positive control shows that the rule detects spill-over when it is there. In simulation, we plant a spatial contrast created only by spill-over at a known fraction, beside a genuine response contrast in the same world, and run the same sweep. The prediction, set before the run, is that the spill-over contrast

breaks at the first grid point at or above the planted fraction, while the response contrast holds across the grid. **[PENDING RUNS: planted spill-over control: planted fraction, observed κ^* , and the response contrast’s status; if κ^* does not track the planted fraction, report it as found and present the breakdown point as a lower-power check]**

Of the spill-over forms the sweep leaves open (Section 2.4), several were refitted on the primary section, three seeds each (Supplement S8.2): per-cell and per-gene spill-over rates move no read by more than 1.3 seed standard deviations, but a fixed false-positive floor lowers the relocation read from 0.65 to 0.50.

4 Discussion

[TODO: discussion]

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